User Requirements Specification (URS) for the Music BoomBoxd



Date	**:**	**04/04/2024**
Version	**:**	**1.0**
State	**:**	**Finished**
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Version History

Version	Date	Author(s)	Changes	State
0.1	03/04/2024	Szymon	Made a beginning on the URS plan.	Finished
1.0	04/04/2024	Szymon	Finished the first version of the URS plan	Finished
1.1	05/04/2024	Szymon	Updated the URS according to small feedback from the WKS teacher	Finished
1.2	07/06/2024	Szymon	Updated the URS according to feedback from the WKS teacher	Finished

Glossary

RBAC	Role-Based Access Control
URS	User Requirements Specification
CRUD	Create Read Update Delete
HTTPS	Hypertext Transfer Protocol Secure

Introduction

The User Requirements Specification (URS) outlines the functional and non-functional requirements for the development of the Music BoomBoxd application. This application aims to provide a platform for music enthusiasts to discover new music, connect with like-minded individuals, and engage in meaningful discussions about songs, albums, and artists.

Functional Requirements

FR.1 User Profiles

1. The system will allow users to create, view, update, and delete their profiles (CRUD).
2. Users will be able to select their 5 favorite albums and 5 favorite songs to display on their profiles.
3. Users will be able to rate albums from 0 to 5.
4. Users with a musician role can add songs and albums.
5. Users with a critic role can add songs and albums and also review them from 0 to 5.

FR.2 Review System

1. Users will be able to rate and see average ratings of songs, albums, and artists.
2. The system shall provide a filter for users to get songs, albums, and artists based on their favourite genre, artist and rating.

FR.3 Content Display and Customization

1. The system shall display the most recent albums.
2. The system shall categorise music by artist, genre, and rating for easy search and discovery.
3. Users will be able to customise their music preferences in the settings.

FR.4 User Categorization

1. The users shall be categorized as Critic, Musician or Regular based on their engagement and expertise
2. Users with a musician role can add songs and albums.
3. Users with a critic role can add songs and albums and also review them from 0 to 5.
4. Regular users will only be able to look at the most recent albums and songs that are on the homepage.

Non-Functional Requirements

NFR.1 Usability

NFR.1.1 The system will have an intuitive and user-friendly interface to facilitate ease of use of the application.
NFR.1.2 Features such as search filters and sorting options shall be provided for enhanced usability.

NFR.2 Scalability

NFR.2.1 The system architecture will be designed to accommodate future growth and scalability requirements, including increased data volume and user concurrency.
NFR.2.2 Modular and extensible components will be utilised to support incremental updates and enhancements to functionality.

NFR.3 Reliability

NFR.3.1 Comprehensive error handling
NFR.3.2 Multiple data validation check
NFR.3.3 Backup and recovery mechanisms shall be implemented to safeguard against data loss or system failures

NFR.4 Security

NFR.4.1 The system will implement role-based access control (RBAC) to control sensitive information and functionalities based on user roles.
NFR.4.2 The system will implement user authentication mechanisms (e.g., username/password, multi-factor authentication). This will help to verify the identity of users and add another layer of security to the system.
NFR.4.3 The system will implement data encryption protocols to protect confidential data during transmission (e.g., https). This will ensure that confidential data is protected.

NFR.5 Performance

NFR.5.1 The system shall handle concurrent user interactions without significant performance degradation.
NFR.5.2 Response times for common operations should be within acceptable limits, even under peak load conditions.

Use Cases

Use Cases for FR.1 User Profiles

UC1. 1 Create Profile

Actor: User
Pre-conditions: None

Main Success Scenario:

1. The User enters the website and selects “Not a member yet?” at the bottom of the login form.
2. The system displays the register form page.
3. The User fills in the required details.
4. The User selects "Register" to register as a new user.
5. The system creates the profile and navigates the user to the login page.
6. The user logs in with the new account.
7. The system navigates the user to the Homepage.

Extensions:

3. If any required fields are missing, the system prompts the user to complete them..
6. If the user encounters an error while logging in to the page, an error message is displayed.

UC1. 2 View Profile

Actor: User
Pre-conditions: Logged in as a registered user

Main Success Scenario:

1. The user selects “Profile” from the menu.
2. The system displays the user's profile information.

Extensions:

2. If the system fails to load the user’s profile and any fields are missing, the system shows an error message and prompts the user to try again.

UC1. 3 Update Profile

Actor: User
Pre-conditions: Logged in as a registered user

Main Success Scenario:

1. The user selects “Profile” from the menu.
2. The system displays the user's profile information.
3. User fills in new username or email or selects a new profile picture.
4. The user selects “Update Profile”.
5. The system updates the profile.

Extensions:

2. If the system fails to load the user’s profile and any fields are missing, the system shows an error message and prompts the user to try again.
5. If the user tries changing the username to something else but that username is already taken by another user an error message shows stating that this username is taken. the system displays an error message.
5. If the user tries changing the email to something else but that email is already taken by another user an error message shows stating that this email is taken. the system displays an error message.
5. If the update does not work. The system shows an error message and displays the most recent user data.

UC1. 4 Delete Profile

Actor: User
Pre-conditions: Logged in as a registered user

Main Success Scenario:

1. The user selects “Profile” from the menu.
2. The system displays the user's profile information.
3. The user selects “Deactivate Account”.
4. The system deactivates the account.
5. The system redirects to the login page.

Extensions:

4. If the user encounters an error while deactivating the account, an error message is displayed.

Use Cases for FR.2 Review System

UC2.1 Rate Content

Actor: User
Pre-conditions: Logged in as a registered critic user.

Main Success Scenario:

1. The user navigates to a song, album, or artist page.
2. The user selects a song or album or artist that they want to rate.
3. The user selects a rating from 0 to 5 stars.
4. The system saves the rating.
5. The system updates the average rating.

Extensions:

4. If an error occurs, the system displays an error message.

UC2.2 View Ratings

Actor: User
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The user navigates to a song, album, or artist page.
2. The user sees the average rating out of 5 stars.

Extensions:

2. If an error occurs when loading the average rating, the system displays an error message and asks to reload the page.

UC2.3 Filter Content by Ratings

Actor: User
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The user navigates to the homepage.
2. The user uses a filter to find songs, albums, or artists based on ratings, genre, or artist.
3. The system shows songs, albums or artists based on rating, genre or artists

Extensions:

3. If no results match, the system displays a “No results found” message.

Use Cases for FR.3 Content Display and Customization

UC3.1 View Recent Albums

Actor: User
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The user navigates to the homepage.
2. The system displays the most recent albums and songs.
3. For more Albums the user goes to the albums page.

Extensions:

2. If an error occurs, the system displays an error message.
3. If an error occurs, the system displays an error message.

UC3.2 Search and Categorize Music

Actor: User
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The user navigates to the homepage.
2. The user uses the search function to find music by ratings, genre, or artist.
3. The system displays categorized results.

Extensions:

2. If no results match, the system displays a “No results found” message.

UC3.3 Customize Music Preferences

Actor: User
Pre-conditions: Logged in as a registered user.

Main Success Scenario:

1. The user selects “Profile” from the menu.
2. The system displays the user's profile information.
3. The user customizes their music preference by changing their favourite genre.

Extensions:

3. If an error occurs, the system displays an error message.

Use Cases for FR.4 User Categorization

UC4.1 Categorize User

Actor: User
Pre-conditions: Logged in as a registered regular user.

Main Success Scenario:

1. The user selects “Profile” from the menu.
2. The system displays the user's profile information.
3. The user selects “Upgrade Account” from the profile page or the menu.
4. The system redirects the user to the upgrade account page with upgrading account information and costs.
5. The user has sufficient funds to upgrade and chooses the category they want to upgrade to.
6. The system updates the user’s role.

Extensions:

5. If an error occurs due to the user not having enough funds to upgrade, the system displays an error message.
6. If an error occurs, the system displays an error message.

UC4.2 Musician Adding Content

Actor: Musician
Pre-conditions: Logged in as a registered musician user.

Main Success Scenario:

1. The Musician selects “Form to add albums” from the menu to add an album.
2. The Musician selects “Form to add songs” from the menu to add a song.
3. The Musician fills in the content for the form.
4. The Musician selects “Save” to add a song or an album.
5. The system saves the content and redirects the Musician to the homepage.
6. The system displays the new content on the homepage.

Extensions:

4. The Musician selects “Cancel” to go back to the homepage..
5. If an error occurs, the system displays an error message.

UC4.3 Critic Adding Content

Actor: Critic
Pre-conditions: Logged in as a registered Critic user.

Main Success Scenario:

7. The Critic selects “Form to add albums” from the menu to add an album.
8. The Critic selects “Form to add songs” from the menu to add a song.
9. The Critic fills in the content for the form.
10. The Critic selects “Save” to add a song or an album.
11. The system saves the content and redirects the Critic to the homepage.
12. The system displays the new content on the homepage.

Extensions:

6. The Critic selects “Cancel” to go back to the homepage..
7. If an error occurs, the system displays an error message.